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Aruna left for city A from city B at 5.20 a.m. She travelled at the speed of 80 km/hr for 2 hours 15 minutes. After that the speed was reduced to 60 km/hr. If the distance between two cities is 350 kms, at what time did Aruna reach city A?

- (a) 9.20 a.m (b) 9.25 a.m.
- (c) 9.35 a.m
- (d) None of these

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The average speed of a train in the onward journey is 25% more than that in the return journey. The train halts for one hour on reaching the destination. The total time taken for the complete to and fro journey is 17 hours, covering a total distance of 800 km. The speed of the train in the onward journey is

- (a) 45 km/hr (b) 47.5 km/hr
(c) 52 km/hr (d) 56.25 km/hr

A car covers a distance of 715 km at a constant speed. If the speed of the car would have been 10 km/hr more, then it would have taken 2 hours less to cover the same distance. What is the original speed of the car?

- (a) 45 km/hr (b) 50 km/hr
(c) 55 km/hr (d) 65 km/hr





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Time gain

A car starts from A and B travelling 20 km an hour. $1\frac{1}{2}$ hours later another car starts from A and travelling at the rate of 30 km an hour reaches B $2\frac{1}{2}$ hours before the first car. Find the distance from A to B.

- (a) 280 km (b) 260 km
(c) 240 km (d) none of these

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A car travels from P to Q at a constant speed. If its speed were increased by 10 km/hr, it would have taken one hour lesser to cover the distance. It would have taken further 45 minutes lesser if the speed was further increased by 10 km/hr. What is the distance between the two cities ?

- (a) 420 km (b) 540 km
(c) 600 km (d) 650 km

A man in a train notices that he can count 21 telephone posts in one minute. If they are known to be 50 metres apart, then at what speed is the train traveling?

- (a) 55 km/hr (b) 57 km/hr
(c) ~~60~~ km/hr (d) 63 km/hr

- 1 A person travels 200 kms in 3 hrs, next 300 kms in 4 hrs, last 500 kms in 3 hrs. What is his average speed?
- A) 120 kmph
 - B) 100 kmph
 - C) 80 kmph
 - D) 130 kmph

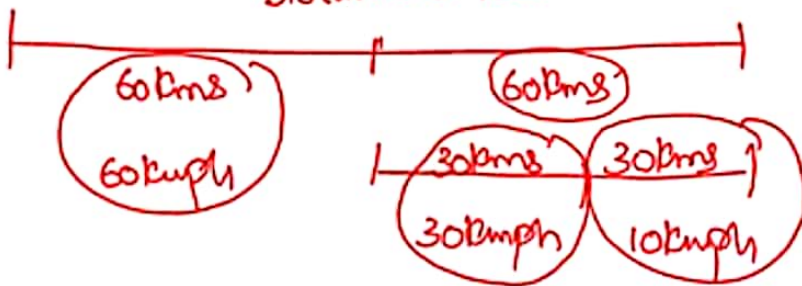
2

A, B and C are on a trip by a car. A drives during the first hour at an average speed of 50 km/hr. B drives during the next 2 hours at an average speed of 48 km/hr. C drives for the next 3 hours at an average speed of 52 km/hr. they reached their destination after exactly 6 hours. Their mean speed was:

- (a) 50 km/hr
- (b) $50 \frac{1}{3}$ km/hr
- (c) 52 km/hr (d) 52 km/hr

(Assume)

Distance = 120 kms



$$A.S = \frac{T.D}{T.T} = \frac{120}{\frac{60}{60} + \frac{30}{30} + \frac{30}{10}}$$

$$= \frac{120}{5} = 24 \text{ km/h}$$

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A tourist covers half of his journey by train at 60 km/h, half of the remainder by bus at 30 km/h and the rest by cycle at 10 km/h. Average speed of the tourist during the journey is (ESE-19)

- (a) 36 km/h
- (b) 33 km/h
- (c) 24 km/h
- (d) 18 km/h

$$T = \frac{D}{S}$$

$$\text{Average Speed} = \frac{T \cdot D}{T \cdot T} = \frac{3D}{T \cdot T}$$



A person divides his total route of journey into three equal parts and decides to travel the three parts at the speeds of 80 kmph, 60 kmph and 30 kmph respectively. What is the average speed during the journey?

- a) 40 kmph
- b) 45 kmph
- c) 48 kmph
- d) 49 kmph

* For a car there are 5 tyres including one spare tyre ($4 + 1$). All tyres are equally used. If the total distance travelled by the car is 40000 km then what is the average distance by the

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- A. 10000 B. 40000
C. 32000 D. 8000



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